

Mammary development in prepubertal gilts fed restrictively or ad libitum in two sub-periods between weaning and puberty

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Abstract

Two experiments were conducted to assess whether feeding level in the period from weaning to puberty can affect the amount of mammary tissue at puberty. Four female pigs from eight (exp. 1) to ten (exp. 2) litters were allocated to four treatment groups within litter. From day 28 to 90 of age (period 1) and from day 90 to slaughter, at approximately 5 1/2 months of age (period 2), the gilts were given either ad libitum or restrictive access to feed in a 2 × 2 factorial arrangement. Neither the amount of dissected mammary tissue nor the amounts of DNA or RNA were affected by feeding regime in period 1. However, ad libitum feeding in period 2 gave rise to more dissected mammary tissue (52%, $P < 0.0001$), RNA (37%, $P = 0.006$) and DNA (28%, $P = 0.005$) compared to restricted feeding. Cyclicity gave rise to increases in mammary DNA ($P = 0.05$) and RNA ($P = 0.02$). There were no clear correlations between mammary measures and plasma IGF-I or IGF-binding proteins. It is concluded that a high feeding level from three months of age to puberty has a stimulatory effect on mammary development in prepubertal gilts.

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1. Introduction

The rate of mammary development in prepubertal gilts varies according to the phase of growth. Indeed, Sørensen et al. (2002) showed that until approximately 90 days of age, accumulation of mammary tissue and

DNA is slow in Landrace × Yorkshire gilts while the rate of accumulation thereafter increases 4 to 6 fold. In dairy heifers, which have a similar pattern of mammary growth as gilts (Sinha and Tucker, 1968), the period with relatively rapid mammary growth is of particular importance because a high feeding level during this period can adversely affect mammary growth and subsequent milk production (Sejrsen and Purup, 1997). In the swine industry, it is essential to maximize sow milk production since milk is the major source of nutrients

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for suckling piglets. Sow milk yield is positively correlated to the amount of mammary tissue (Nielsen et al., 2001), and it is therefore important that the feeding regime of gilts be designed to support mammary growth and development in order to preserve the genetic potential for subsequent milk production. The growth factor IGF-I and its binding proteins are among the factors that regulate mammary growth and development (Wood et al., 2000), yet their possible role in mammaryogenesis in swine is not known.

Based on the apparent change in rate of mammary growth at 90 days of age in gilts (Sørensen et al., 2002), the main purpose of the present work was to investigate whether high or moderate feeding level before and/or after 90 days of age affects mammary development in swine. In addition, plasma concentrations of IGF-I and IGF-binding proteins were determined to establish their possible involvement in mammaryogenesis.

2. Material and methods

Two experiments were conducted, exp. 2 (10 litters) being a repetition of exp. 1 (8 litters). In both experiments, litters of four female pigs (Landrace × Yorkshire) were allocated to four treatment groups within litter, ensuring that average body weight was similar across treatments. The experiments commenced after weaning at 28 days of age, at which time weight of piglets were 8.1 kg (SD=1.4). From day 28 to 90 (period 1) and from day 90 to slaughter (period 2), pigs were given either ad libitum (A) or restricted (R) access to feed in a 2 × 2 factorial arrangement, i.e. ad libitum in periods 1 and 2 (AA), ad libitum in period 1 and restricted in period 2 (AR), restricted in period 1 and ad libitum in period 2 (RA), and restricted in periods 1 and 2 (RR). Feed allowance of the restrictively fed animals is outlined in Table 1. From day 28 to 56, all pigs were offered a weaner diet containing 14.19 MJ ME, 242 g CP and 14.8 g lysine per kg of feed, from day 57 to 84, a grower diet (Grower-1) containing 13.98 MJ ME, 230 g CP and 12.9 g lysine per kg of feed, from day 85 to

112, a grower diet (Grower-2) containing 13.65 MJ ME, 199 g CP and 10.3 g lysine per kg of feed, and from day 113 to slaughter, a finisher diet containing 13.79 MJ ME, 181 g CP and 9.2 g lysine per kg of feed. For details on diet composition see Kristensen et al. (2002). The pigs were raised individually in pens with partly slatted floors. Chopped straw was used as bedding. Feed intake was recorded individually and pigs were weighed every two weeks.

Jugular blood samples were obtained in the last week of period 1 and in the last week before slaughter. Plasma samples were analysed for IGF-I (Frystyk et al., 1995) by non-competitive time-resolved immunofluorimetry. IGFBPs were analysed by Western ligand blotting using the method of Hossenlopp et al. (1986), as described by Vestergaard et al. (1995). In agreement with Donovan et al. (1994), five MW_r bands were detected on the autoradiographs and corresponded to the doublets for IGFBP-3 (39–43 kD), IGFBP-2 (34-kD), a 28 kD band probably corresponding to IGFBP-5, IGFBP-1 and/or glycosylated IGFBP-4, and a 24 kD band probably corresponding to IGFBP-4. In this paper we only report data for IGFBP-3, IGFBP-2 and total IGFBP. The bands on the autoradiographs were quantified by laser-densitometry (exp. 1) or desktop scanning densitometry (exp. 2), and the relative abundance of the IGFBPs was expressed in arbitrary units.

The pigs were slaughtered by litter group at the slaughterhouse of Research Centre Foulum, after CO₂ stunning at 171 (SD=3.6) or 162 (SD=2.6) days of age in experiments 1 and 2, respectively. After scalding, the udder and the first muscle layer dorsal to the udder were excised from the carcass, split in left and right halves and frozen at –20°C. The ovaries were examined for the presence of corpora lutea. In exp.1, four glands (numbers 2–5) from one half, and in exp. 2, all glands except the front and rear glands from both halves were dissected. The udder halves were partly thawed and cut into 1.0 cm slices. Mammary tissue (parenchyma and mammary fat pad) was then separated from adjacent tissue (skin, fat and muscle), weighed and stored at –20 °C. A few udder halves had 8 glands while the rest had 7, thus

Table 1
Feed allowance for restrictively fed animals

Age, weeks	4	5	6	7	8	9	10	11	12	13	14	15	16	17→
Feed, kg/day	0.15	0.3	0.45	0.6	0.75	0.9	1.1	1.3	1.5	1.7	1.9	2.1	2.3	2.5

Table 2

Effect of feeding in period 1 (28 to 90 days of age) on performance during that time period

Period 1	Ad libitum	Restricted	P-value	RMSE ^a
Weight, kg (day 28)	8.0	8.2	>0.2	0.9
Feed intake, kg/day	1.15	0.76	<0.0001	0.09
Gain, g/day	648	517	<0.0001	59
Kg feed/kg gain	1.81	1.47	<0.0001	0.13

^a Root mean square error.

the recorded weights were adjusted to 7 glands per udder half. The frozen mammary tissue was cut into small pieces, minced in a meat grinder and mixed thoroughly. A sample of the minced tissue was used for measurements of total RNA and DNA, as described by Oksbjerg et al. (2000), for exp. 1, or with small modifications (see Therkildsen et al., 2002) for exp. 2. Total amount of DNA in the mammary tissue was calculated and used as an indicator of total number of mammary cells since the amount of DNA is constant at approximately 6 pictograms per cell (Lehninger, 1975).

All statistical analyses were conducted by means of procedures and options in SAS (SAS Inst. Inc., 1999–2000). Data from the two experiments were pooled unless otherwise stated. Data from period 1 were analysed in a model including the fixed effects of litter and feeding treatment in period 1. Data from period 2 including the mammary gland data were analysed in a model including the fixed effects of litter, feeding treatment in period 1, feeding treatment in period 2, and the interaction between feeding treatments in periods 1 and 2. Systemic differences between exp. 1 and exp. 2 are included in the litter effect and will thus be corrected for by this effect. Two data sets were constructed for the mammary gland data. In one data set, data from all animals that had not attained puberty were included and analysed according to the model for period 2 data. In

another data set, data from experiment 1 were analysed according to the model for period 2 data with addition of the effect of cyclicity since eight animals had attained puberty before slaughter. All results are presented as least squares means. Correlation coefficients between the mammary data and the IGF-I and IGF-BP measures were calculated by the MANOVA option.

3. Results

One pig in each of the two experiments was excluded because of illness and death. The performance of animals is shown in Tables 2 and 3. In period 1, pigs fed ad libitum consumed 52% more feed, grew 25% faster and consumed 24% more feed per kg of gain than restrictively fed pigs (Table 2). In period 2, pigs fed ad libitum (AA and RA) consumed 33% more feed, grew 27% faster and consumed 7% more feed per unit gain than pigs fed restrictively (AR and RR). There were no interactions between feeding regime in period 1 and feeding regime in period 2 (Table 3).

The effects of feeding regime on mammary tissue and nucleotides are shown in Table 4. Neither the amount of dissected mammary tissue nor the concentrations or amounts of RNA and DNA were affected by feeding regime in period 1. However, in period 2, ad libitum feeding gave rise to 52% more dissected mammary tissue ($P < 0.0001$). Tissue concentrations of RNA and DNA were generally highest in pigs fed restrictively in period 2, but only the DNA concentration reached a probability level less than 0.05. Ad libitum feeding in period 2 gave rise to 37% more mammary RNA ($P = 0.006$) and 28% more mammary DNA ($P = 0.005$) than restricted feeding. There were no significant interactions between feeding regime in periods 1 and 2 on mammary data.

Table 3

Effect of feeding in periods 1 (28 to 90 days of age) and 2 (90 days of age to slaughter) on performance in period 2

Period 1	Ad libitum (A)		Restricted (R)		Effect of feeding (P-value)			RMSE ^a
	A	R	A	R	Period 1	Period 2	Per 1 × per 2	
Period 2								
Gain, g/day	1000	795	1075	844	0.006	<0.0001	0.56	89
Feed intake, kg/day	3.05	2.23	3.09	2.27	0.55	<0.0001	0.97	0.24
Kg feed/kg gain	3.04	2.82	2.86	2.69	0.003	0.0004	0.56	0.20
Weight at slaughter, kg	125	108	122	105	0.09	<0.0001	0.85	8

^a Root mean square error.

Table 4

Effect of feeding in periods 1 (28 to 90 days of age) and 2 (90 days of age to slaughter) on amount of mammary tissue, RNA and DNA

Period 1	Ad libitum (A)		Restricted (R)		Effect of feeding (<i>P</i> -value)			RMSE ^a
Period 2	A	R	A	R	Period 1	Period 2	Per 1 × per 2	
Tissue, g/gland	85.6	54.1	77.5	53.2	0.21	<0.0001	0.33	12.8
RNA, µg/g tissue	474	540	490	521	0.96	0.22	0.67	140
DNA, µg/g tissue	290	364	345	381	0.15	0.03	0.46	90
RNA, mg/gland	40.2	29.6	37.9	27.6	0.54	0.006	0.96	12.8
DNA, mg/gland	24.8	19.6	26.3	20.3	0.56	0.005	0.84	6.6

^a Root mean square error.

Eight of the 31 gilts in exp. 1 had attained puberty at slaughter, as judged by the presence of corpora lutea on the ovaries, but there was no clear relation to feeding regime since four were from the AA, one from the RA, and three from the RR group. The amount of mammary tissue and the concentration of nucleotides were not significantly affected by cyclicity, but the trends were towards highest values in the cycling gilts resulting in 64% more RNA ($P=0.02$) and 47% more DNA ($P=0.05$) in cycling compared to non-cycling gilts (Table 5).

Circulating IGF-I and IGFBP data are shown in Tables 6 and 7. At the end of period 1, IGF-I and IGFBP-3 were greater whereas IGFBP-2 was lower, in ad libitum compared to restrictively fed pigs (Table 6). At the end of period 2, IGFBP-2 was lower in ad libitum compared to restrictively fed pigs, whereas there were no clear trends for IGF-I and IGFBP-3 concentrations. There were no interactions between feeding regime in periods 1 and 2 on these variables (Table 7).

The plasma IGF-I and IGFBP measures showed no correlations with mammary tissue measures (data not shown), except for weak positive correlations between circulating IGF-I in the sample taken before slaughter and concentration and amount of RNA in mammary tissue ($r=0.29$ and 0.32 , respectively, $P<0.10$).

4. Discussion

The growth performance of the gilts confirms the expected effects of the applied feeding regimes, with ad libitum-fed animals growing faster than restrictively fed animals due to a greater feed intake. Performance in consequence of the applied feeding regimes is discussed in detail by Oksbjerg et al. (2002).

Our study shows that even before puberty, mammary development can be affected by feeding regime, i.e. ad libitum feeding in a period before attainment of puberty enhances mammary development. This finding is contrary to what was found in cattle (Sejrsen and Purup, 1997), but in agreement with previous findings in pigs. Farmer et al. (2004) found that dietary protein level from 90 days of age onward did not affect mammary development whereas ad libitum feeding gave rise to more parenchymal as well as extra-parenchymal mammary tissue compared to restrictive feeding. In the present experiments, parenchymal and extra-parenchymal mammary tissue were not separated, but total mammary tissue increased by 52% in ad libitum compared to restrictively fed gilts. This is in agreement with the 64% increase in total mammary tissue found by Farmer et al. (2004) for similar treatments from 90 to approximately 200 days of age. Also a study by Lyvers-Peffer

Table 5

Effect of cyclicity on amount of mammary tissue, RNA and DNA

Period 2	Cycling (<i>n</i> =8)	Non-cycling (<i>n</i> =23)	<i>P</i> -value	RMSE ^a
Tissue, g/gland	68.3	59.9	0.36	13.2
RNA, µg/g tissue	651	477	0.13	159
DNA, µg/g tissue	381	310	0.29	97
RNA, mg/gland	44.3	27.0	0.02	10.0
DNA mg/gland	26.1	17.7	0.05	6.0

^a Root mean square error.

Table 6

Effect of feeding in period 1 (28 to 90 days of age) on IGF-I and IGFBP^a in plasma harvested at the end of period 1

Period 2	Ad libitum	Restricted	<i>P</i> -value	RMSE ^b
IGF-I, ng/ml	266	193	0.001	66
IGFBP-2	216	295	0.01	88
IGFBP-3	1013	854	0.02	229

^a Arbitrary units.^b Root mean square error.

and Rozeboom (2001) supports our finding that restrictive feeding before puberty may decrease mammary development. They found that gilts fed moderately in two periods before puberty (9–12 and 15–20 weeks of age) had decreased mammary DNA in late pregnancy compared to gilts fed more generously.

In non-pregnant gilts (200 days of age), there is approximately four times more extra-parenchymal than parenchymal tissue, and when gilts are restricted-fed the accumulation of extra-parenchymal tissue is reduced to a greater extent than that of parenchymal tissue (Farmer et al., 2004). This may be important for an optimal development of the mammary gland because mammary ducts grow and branch into mammary adipose tissue, which is the main component of the extra-parenchymal tissue. Thus, when there is less extra-parenchymal tissue in restrictively fed compared to ad libitum-fed gilts, there is less adipose tissue for the mammary ducts to grow and branch into. The optimal amount of mammary adipose tissue required for mammary development is not known. However, our data indicate that ad libitum feeding from three months of age is sufficient since starting the period of ad libitum feeding at weaning did not give rise to more mammary tissue at puberty. Our data also show that ad libitum feeding only in the period from weaning to three months of age has no significant effect on the amount of mammary tissue present. Thus it seems that ad libitum feeding should be practised in a period up to the attainment of puberty. The present experiments were not designed to determine the optimal duration of the period with ad libitum feeding, and it is not known whether the onset of the period of ad libitum feeding should be later than 90 days of age. Likewise, it is not known whether the period of ad libitum feeding should be extended to the time of service for gilts that are not serviced at the first

oestrus. However, results of Farmer et al. (2004) suggest that ad libitum feeding may be extended into the period of cyclicity with beneficial effects on mammary development. Unlike the situation in cattle (Sejrsen and Purup, 1997), ad libitum feeding in the entire prepubertal period does not seem to have detrimental effects on mammary development in swine. The reason for this species difference is not known, but it underlines the fact that results on the impact of nutritional regime on mammary development cannot be extrapolated from one species to another.

Cyclicity clearly increased the amount of nucleotides in mammary tissue, yet this increase occurred without a significant rise in the amount of mammary tissue, suggesting extensive growth of mammary ducts into the mammary adipose tissue after the attainment of puberty. These results are in agreement with those of Farmer et al. (2004) who also found that cyclicity stimulated mammary development. Knowing the importance of estrogen for mammary development in pregnant swine (Kensinger et al., 1986) it can be hypothesised that hormones originating from the ovaries are responsible for this cyclicity-stimulated development.

Evidence for a role of the IGF signalling pathway in mammary epithelial growth originates from in vitro work in laboratory animals (Imagawa et al., 1986; Deeks et al., 1988) and in cattle (Purup et al., 2000a) as well as work with transgenic mice (Hadsell and Bonnette, 2000). It was shown that IGF is expressed in mammary tissue (Walden et al., 1998) and it is likely that locally synthesised IGF is important for stimulation of mammary tissue growth (Purup et al., 2000b; Weber et al., 2000). Indeed, Berry et al. (2003) reported that the decrease in mammary development associated with ovariectomy in prepubertal heifers is associated with decreased IGF-I mRNA expression within the

Table 7

Effect of feeding in periods 1 (28 to 90 days of age) and 2 (90 days of age to slaughter) on IGF-I and IGFBP^a in plasma collected approximately one week before slaughter

Period 1	Ad libitum (A)		Restricted (R)		Effect of feeding (<i>P</i> -value)			RMSE ^b
Period 2	A	R	A	R	Period 1	Period 2	Per 1 × per 2	
IGF-I, ng/ml	217	194	203	194	0.64	0.27	0.64	53
IGFBP-2	192	284	201	253	0.60	0.01	0.37	77
IGFBP-3	817	787	842	808	0.72	0.62	0.98	230

^a Arbitrary units.

^b Root mean square error.

mammary gland but that no changes were observed in systemic concentrations of IGF-I. Therefore, systemic IGF may be of relatively minor significance for mammary growth. Flint et al. (2000) also stated that the IGFBPs may have biological effects that are independent of circulating IGF concentrations. All these findings are in agreement with the lack of clear correlation coefficients found between the mammary tissue measures and the plasma IGF-I and IGFBPs measures in the present experiments.

There are only a few reports in the literature on the effect of feeding strategy before puberty on subsequent milk production of sows. Feeding level of prepubertal gilts in the entire period from around weaning to puberty (Kirchgeßner et al., 1984; Sørensen et al., 1998; Le Cozler et al., 1998) or from 75 days of age to puberty (Le Cozler et al., 1999) showed no clear effects on subsequent milk production, although Kirchgeßner et al. (1984) found that piglets from sows that were previously restricted-fed had the highest growth rate in parity two, but not in parities one or three. Lyvers-Peffer and Rozeboom (2001) found somewhat surprisingly that gilts with increased mammary DNA in late pregnancy after generous feeding before puberty weaned smaller litters compared to gilts fed more moderate before puberty. They speculate that this may be because the generously reared gilts consumed less feed during lactation. Thus, further research is warranted to clarify whether differences in feeding and mammary development before puberty will give rise to differences in subsequent milk yield.

In summary, a high feeding level from three months of age to puberty has a stimulating effect on mammary growth and development in prepubertal gilts.

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